

Java Memory Management

One of the Java languages finest achievements is its memory management. Java has built in garbage collection. Developers can create objects without worrying about explicitly allocating and de-allocating memory.

The garbage collector automatically reclaims memory it deems not in use. This eliminates memory leaks and other memory related problems. Garbage collection maintains program integrity and is an important part of Java's security strategy. It also stops heap fragmentation which occurs when memory is manually de-allocated freeing up a lot of intermittent space in between.

Learning tools

The tools available for learning and programming in java are excellent. Not to mention free. Sophisticated Integrated Development Environments (IDEs) such as NetBeans and Eclipse as well as web servers (TomCat), application servers (Glassfish, JBoss).

Software

The Oracle docs have specified a set of highly useful tools to learn Java and Java-like languages depending on both age and experience. Whether you are an experienced programmer or a complete novice there is something in there for you. Whether you are 5 years old or 80 they have an easy interface for you to begin with.

Scratch: A simple programming interface with drag and drop features. This tool was built for kids between the ages of 5-15 with no programming experience.

Greenfoot: A 2D visual software with a built-in code editor to create games and simulations. Again, this is for starters with no prior programming experience, but aged 13 upwards.

Alice: A 3D educational tool for animations with a drag and drop interface. No programming experience required. No age limit either.

BlueJ: This is a slightly more sophisticated tool where users are expected to write their own code. It is not an overwhelming IDE like Eclipse or NetBeans and provides the bare basics to run a Java program. A little coding experience is required to use BlueJ. High school kids usually start with this.

NetBeans/ Eclipse: These are fully featured Java Integrated Development Environments used in educational as well as professional environments.